From Sparse Satellite Images to Continuous Visual Sequence: A Collaborative Diffusion Framework for Internal Solitary Waves

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***Abstract*—Internal solitary waves (ISWs) are a significant oceanic phenomenon, particularly in the South China Sea (SCS). The propagation of ISWs influences the transport of matter, energy, and momentum within the ocean. They also pose a considerable risk to offshore engineering and submarine safety. To leverage the advantages of ISWs and to mitigate the associated risks to offshore activities, it is valuable to understand their continuous propagation process over a period of time. At present, there is a lack of tools capable of illustrating the continuous propagation of ISWs. Consequently, reconstructing the continuous propagation process of ISWs based on sparse remote sensing imagery is a valuable endeavor. To address this issue, this article defines the completion of the ISW propagation process as an end-to-end visual generation task and proposes a novel collaborative diffusion framework that directly generates high-fidelity visual sequences of ISW propagation. The core innovation of the framework is the disentanglement of static appearance and dynamic motion, together with the integration of physical information into both representations. It employs a dual-path architecture where an appearance model learns the static morphology of ISWs from a single image, while a motion model learns their propagation kinematics. These two components are synergistically constrained by a shared-weights mechanism and physical information, ensuring a physically coherent spatiotemporal representation. The proposed framework is validated using numerical simulation data, synthetic aperture radar (SAR) imagery, and geostationary high-speed imager (GHI) imagery. The framework is assessed primarily by visual metrics, supplemented by quantitative metrics where continuous sequences are available. Quantitative results on both GHI and numerical simulations demonstrate the high fidelity of the proposed method. For the GHI data, the errors in propagation velocity remain within 0.7 m/s, while the predicted propagation direction is highly consistent with the observations. For the numerical simulations, the velocity error is only 0.1 m/s, with a direction deviation of 11°. The results confirm that this framework provides a novel and effective tool for reconstructing the propagation process of ISWs over time from otherwise sparse observations.**

***Index Terms*—Diffusion model, internal solitary waves (ISWs), multimodal, remote sensing, visual observation.**

I. Introduction

INTERNAL solitary waves (ISWs) are powerful, large-amplitude nonlinear waves that are widely distributed in the stratified global oceans [1], [2], [3], [4], [5]. As they propagate, they exert a significant influence on the transport of mass, energy, and momentum within the ocean. They also pose a considerable risk to both marine engineering projects and the safety of underwater navigation [6], [7]. Therefore, given their profound oceanic impacts and operational risks, the ability to characterize ISWs and visualize their continuous propagation over a period of time is indispensable for both scientific advancement and maritime safety.

Satellite remote sensing has played an indispensable role in advancing the study of ISWs [8], [9], [10]. Extensive research leveraging multisource satellite imagery (e.g., synthetic aperture radar (SAR) and optical imagery) has successfully mapped ISW generation sites [11], tracked their propagation pathways, and revealed their extreme physical properties, such as amplitudes reaching 240 m [12], [13], [14], [15]. This research paradigm, pioneered in early studies of regions like the Bay of Biscay and the Iberian Peninsula [16], [17], has provided a foundational understanding of ISW dynamics globally. Yet, despite this deep body of knowledge and technological sophistication, a fundamental and unresolved conflict plagues these remote sensing systems: the inherent tradeoff between spatial coverage and temporal resolution. Polar-orbiting satellites, which provide the wide-swath imagery necessary to capture large-scale wave packets, are constrained by long revisit cycles ranging from days to weeks. Although tandem satellite missions have been employed to capture short-term propagation dynamics by observing the same ISW packet twice within a short interval [18], such observations result

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in highly fragmented, discontinuous data. This observational dilemma creates a severe data scarcity and fragmentation problem, which stands as the primary bottleneck hindering a continuous understanding of ISW dynamics. As a result, visually generating the continuous propagation of ISWs over a period of time from these sparse inputs emerges as a critical, yet unresolved, challenge.

The challenge of data scarcity plagues existing approaches designed for ISW research as well. The research trajectory in this area reveals a clear evolution from classic empirical models to modern physics-AI hybrid frameworks. However, despite their continuous advancements, these methods are all fundamentally constrained by the sparse nature of the input data. This limitation prevents them from producing the continuous, high-fidelity visual sequences necessary for a comprehensive observational understanding.

Much of the existing research leveraging these datasets has focused on predicting ISW location. This established paradigm typically involves predicting discrete physical parameters, such as propagation speed [19], [20], and subsequently inferring a future location [21]. In contrast, this article addresses a different, yet largely unexplored, challenge: visual generation. Rather than predicting future states, our work focuses on reconstructing a continuous, high-fidelity visual record of the ISW propagation process from the available sparse observations.

In summary, while artificial intelligence has achieved significant progress in both ISW detection or segmentation and propagation prediction, two critical bottlenecks remain to be overcome as follows.

1) The performance of all existing methods, whether for segmentation or prediction, is fundamentally limited by the inherent "spatiotemporal discontinuity" of satellite remote sensing data. No effective method currently exists to fundamentally address the challenges posed by this data scarcity.

2) In response to this data gap, existing research has primarily focused on the challenge of prediction. In contrast, the separate and more fundamental challenge of visual generation—that is, generating a continuous visual record from the sparse data itself—remains largely unexplored.

Therefore, this article proposes that the key to future breakthroughs lies in a paradigm shift: the development of a new method for end-to-end spatiotemporal visual generation that operates directly on sparse and discontinuous image sequences. Based on the above considerations, from the cutting-edge perspective of generative models [22], this article proposes a collaborative diffusion framework, designed to directly generate spatiotemporally continuous visual sequences from sparse ISW observations. The core of this framework is an innovative dual-network collaborative mechanism: a "feature diffusion network" ensures the physical morphological fidelity of the generated waveforms, while a "trajectory diffusion network" guarantees the kinematic plausibility of their propagation. During the generation process, these two networks are mutually constrained by a shared-weights mechanism and unified metadata embeddings, ultimately enabling the reconstruction of a complete, physically plausible, and dynamically fluid propagation process over a period of time from extremely sparse inputs. This fundamentally overcomes the constraints imposed by data scarcity on continuous ISW observation and analysis.

The article is structured in the following way. Section II presents the related work on ISWs. Section III introduces the research region and data; Section IV outlines the model's structure; Section V covers the experimental results and discussion; and Section VI offers the conclusion.

II. Related Work

Ocean remote sensing has entered the era of big data [5], posing significant challenges for the accurate, efficient, and intelligent extraction of valuable information from massive datasets. Currently, there is a scarcity of continuous data. How to generate continuous observations from the existing discontinuous data? Deep learning makes this a possibility. Deep learning, as a key technology to address this challenge, has demonstrated significant advantages over traditional physics-based or statistical methods in image information processing and is increasingly becoming a major driving force in the field of ocean remote sensing [23].

Accurately detecting and segmenting ISWs from remote sensing imagery is the primary step toward understanding their physical characteristics. Early research, beginning in the 1990s, primarily relied on traditional image processing techniques. For instance, researchers extensively utilized methods such as wavelet analysis [24], [25], Fourier transforms [26], and edge detection [27], [28] for the automatic or semi-automatic identification of ISWs in SAR images. While these methods proved effective in specific scenarios, they generally suffer from two major limitations: first, they require tedious manual parameter tuning for specific sensors and environmental conditions; second, their computational inefficiency makes them unsuitable for the large-scale, automated processing of terabytes of remote sensing data acquired daily.

To address the bottlenecks of traditional methods, a new paradigm represented by deep learning has emerged. In the task of ISW detection and segmentation, researchers have successively proposed models based on Faster R-CNN for object detection [29], U-Net for segmentation, and other advanced architectures like SegNet [30], a two-stage segmentation algorithm [31], and IWE-Net [32] to enhance segmentation accuracy. These methods perform exceptionally well on high-quality image samples. However, current deep learning detection models still face challenges: on one hand, high-precision models such as those based on attention mechanisms often entail high computational costs and a dependency on large-scale, high-quality datasets [33]; on the other hand, the performance of existing models degrades significantly when processing the low-quality, noisy, or cloud-obscured imagery common in real-world scenarios. Therefore, enhancing model robustness in complex, real-world environments—for example, through denoising preprocessing [34] or by developing more powerful generative segmentation models [35] is a pressing issue to be addressed in the field.

Beyond detection and identification, predicting ISW propagation presents a more formidable challenge. Existing

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prediction methodologies have undergone an evolutionary process from empirical-driven approaches to physics-data fusion. Initially, Jackson [36] established a classic empirical model. Subsequently, methods utilizing time-lagged satellite image pairs, such as sun glint imagery from moderate-resolution imaging spectroradiometer (MODIS) and visible infrared imaging radiometer suite (VIIRS), were developed to directly track ISW packets and estimate their phase speeds based on spatial displacement [37]. Subsequently, researchers attempted to incorporate more physical constraints, such as the speed-matrix model proposed by Shen et al. [38]. With the advent of artificial intelligence, data-driven prediction models became mainstream. For instance, Zhang et al. [39], [40], [41] successfully constructed propagation speed prediction models for specific regions like the Sulu-Celebes and Andaman Seas using machine learning and clustering algorithms, effectively addressing the problem of imbalanced sample distributions. More recently, researchers have deeply integrated physical equations with artificial intelligence, proposing advanced physics-AI hybrid prediction models [42] that have further enhanced the accuracy and interpretability of the predictions.

While this body of work shows significant progress, it also highlights fundamental gaps that this article seeks to address: the research community has largely focused its efforts on the prediction challenge, as detailed above. In contrast, the distinct and more fundamental challenge of visual generation—that is, generating a continuous visual record from the sparse data itself—remains largely unexplored.

In the broader field of computer vision, the transition from static image synthesis to continuous visual sequence generation has been heavily catalyzed by the emergence of video diffusion models (VDMs) [43], [44]. Following the unprecedented success of diffusion models in the 2-D domain, the generative landscape has rapidly expanded to video generation tasks [45]. Unlike image generation, synthesizing continuous video sequences introduces the critical challenge of maintaining temporal consistency across high-dimensional frames [46]. To address this, mainstream VDMs typically build upon pretrained 2-D spatial architectures by inserting pseudo-3-D temporal modules, such as 1-D temporal convolutions and temporal attention layers.

While these approaches exhibit strong generative priors in the general domain, directly applying them to complex, physics-driven ocean remote sensing data presents unique difficulties. General-domain VDMs often struggle to preserve both the strict morphological fidelity of marine phenomena and their dynamic propagation trajectories simultaneously. Therefore, how to effectively adapt these cutting-edge generative models to the stringent physical constraints of ISWs remains an open question, further emphasizing the need for a specialized framework to reconstruct continuous propagation processes from sparse satellite observations.

III. Research Region and Data

A. Research Region

As shown in Fig. 1, the primary study area for this research is the northeastern South China Sea (SCS), a region globally

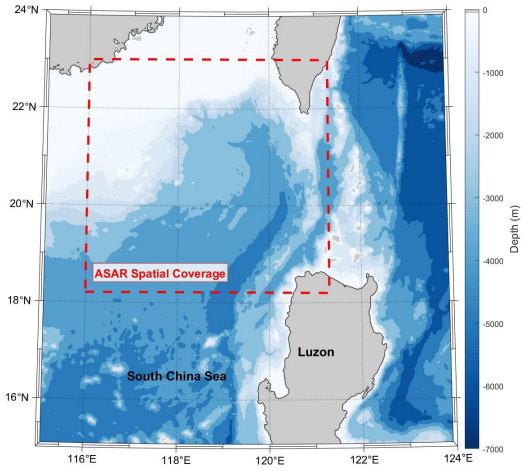


Fig. 1. Bathymetric map of the Luzon Strait region in the SCS. The gray areas represent land, while the blue shading indicates water depth. The color bar unit is meters.

recognized for its frequent and energetic ISW activity [47]. This area, which includes the Luzon Strait and the broad continental shelf, features complex submarine topography and is subject to strong tidal forcing. This combination generates a diverse spectrum of ISW phenomena, with observed wave amplitudes often exceeding 100 m and propagation speeds surpassing 2.5 m/s [48]. Notably, when ISWs encounter obstacles like the Dongsha Atoll, they undergo complex refraction and diffraction processes, creating distinct bifurcated wave patterns [49]. The inherent diversity and intensity of ISW events make the northeastern SCS an ideal natural laboratory for testing the generalization capability and robustness of the proposed generative model against a wide range of wave morphologies and propagation patterns. Recent studies utilizing the Surface Water and Ocean Topography (SWOT) mission have further revealed the abundance of fine-scale processes and ISWs in this region, providing new insights into their spatial distribution [50]. Fig. 2 illustrates examples of the three primary data sources used in this study: numerical simulation, SAR, and geostationary high-speed imager (GHI) imagery.

B. Numerical Simulation Data

The numerical simulation data used in this study were generated using the Coastal and Regional Ocean Community Model (CROCO), which was employed to conduct high-resolution simulations of ISWs in the northeastern SCS and Luzon Strait regions. The model domain spans from 116°E to 124°E and from 16°N to 23°N, with a horizontal resolution of approximately $1/240^\circ \times 1/120^\circ$, corresponding to a grid of 1920×892 points. In the vertical direction, 45 terrain-following layers were configured to accurately resolve the thermocline and the fine-scale vertical structure of nonlinear ISWs. Initial conditions and open boundary forcing were derived from the HYCOM GLBv0.08 global ocean reanalysis dataset. Daily averaged temperature, salinity, sea surface height, and velocity fields were interpolated onto the CROCO grid and used to initialize and drive the simulation. To generate

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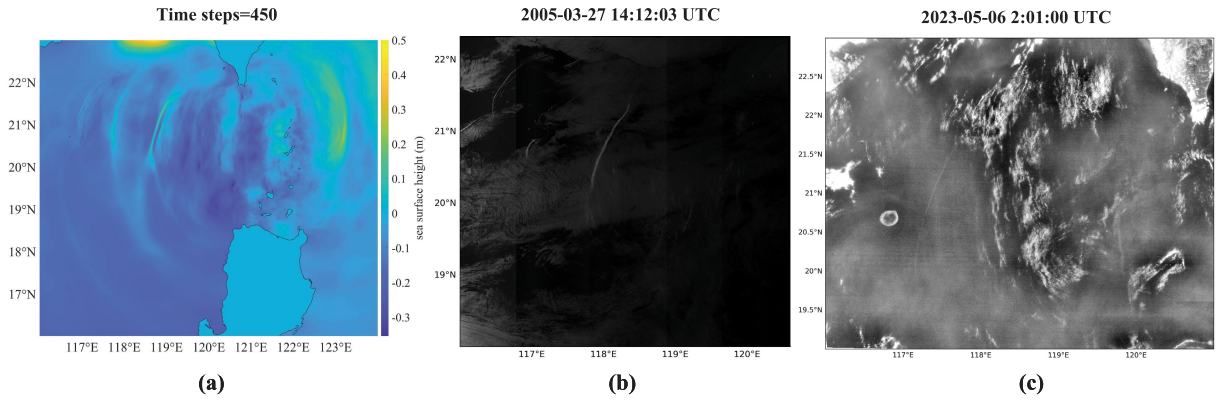


Fig. 2. Multi-source observations of internal solitary waves near the Luzon Strait. (a) Shows numerical model output of ISW generation near the Luzon Strait, where colors indicates sea surface height. (b) Shows ISWs observed in SAR. (c) Shows ISWs captured by the GHI on the FY-4B.

internal tides, four principal tidal constituents (M2, S2, K1, and O1) were extracted from the TPXO7.2 global tidal model and applied to sea surface elevation and barotropic momentum at the open boundaries. All four lateral boundaries were set as open, with sea level governed by standard radiation conditions, barotropic momentum by Flather conditions, and 3-D velocity and tracer fields by a radiation-relaxation scheme. To suppress wave reflection and ensure numerical stability, sponge layers of 30-km width were applied near all open boundaries. The model was integrated for a total of 30 days, using a time step of 60 s. Outputs were saved every 10 min.

To avoid transient effects during the model spin-up phase, we selected data from simulation days 7 to 12, comprising 1009 time steps, as the core dataset for the numerical experiments. These data were also used as input to the motion model in experiments involving discontinuous satellite observations. To ensure the reliability of the simulated ISWs, our CROCO configuration follows established protocols that have been rigorously validated for the SCS [51], [52], [53].

C. Sar Data

This study collected more than one hundred remote sensing images of ISWs in the SCS during the period from 2003 to 2012. The dataset was acquired by the advanced SAR (ASAR) onboard the environmental satellite (ENVISAT). The ASAR sensor operates in the C-band frequency, characterized by a central wavelength of approximately 5.6 cm, and observes ocean surfaces primarily through dual-polarization (VV or HH) data in the wide swath mode (WSM). Technically, the WSM imaging mode provides a spatial resolution of approximately 150 m and a swath coverage of up to 400 km per scene, demonstrating significant advantages for large-scale, high-resolution ocean observations. This type of data lacks short-term and continuous observations. Generating continuous visual sequences to depict their propagation and analyzing their propagation trajectories constitute an important part of the experiments in this study.

D. GHI Data

The GHI optical imagery utilized in this study was acquired from the Fengyun-4B (FY-4B) satellite. GHI integrates a panchromatic band and multispectral channels spanning from 0.445 to 12.5 μm, with spatial resolutions ranging between 0.25 and 4 km. Notably, it is capable of ultrahigh-frequency observation, covering an area of approximately 2000 $\times$ 2000 km within 1 min, facilitating short-term, continuous monitoring of ISWs. The GHI data analyzed herein represent the FY-4B Level-1 product, characterized by a spatial resolution of 250 m and a dynamically variable observation region. During data acquisition, the satellite nadir point was fixed at 133.0°E. A total of 281 GHI images were obtained, spanning from 02:01:00 to 09:01:59 (UTC) on May 6, 2023. This unique, high-frequency GHI image sequence from May 6, 2023, provides a critical real-world test case [54]. It enables the proposed framework to conduct experiments with real and continuous data, allowing for a rigorous validation of the proposed model's performance in a dynamic, observational scenario.

IV. Collaborative Diffusion Framework Description

A. Framework Structure

This article proposes a novel framework that formulates the reconstruction of ISW propagation as an end-to-end visual sequence generation problem. The central hypothesis is that effective visual generation requires disentangling the static visual appearance of ISWs from their dynamic motion. To achieve this, we introduce a collaborative diffusion framework, as illustrated in Fig. 3. The framework's overall input consists of four distinct data modalities as follows.

1) *Image:* The input image is preprocessed and utilized as an initial input for the ISWs feature diffusion model.
2) *Video:* The input video is preprocessed and utilized as an initial input for the ISWs trajectory diffusion model.
3) *Text:* Textual descriptions corresponding to the images and videos are processed to serve as conditional guidance for both the ISWs feature and trajectory diffusion models.
4) *ISW Parameters:* ISW-related parameters, including location, time, temperature, background field flow velocity, and peak-to-peak distance. These parameters are processed through a unified metadata embedding layer

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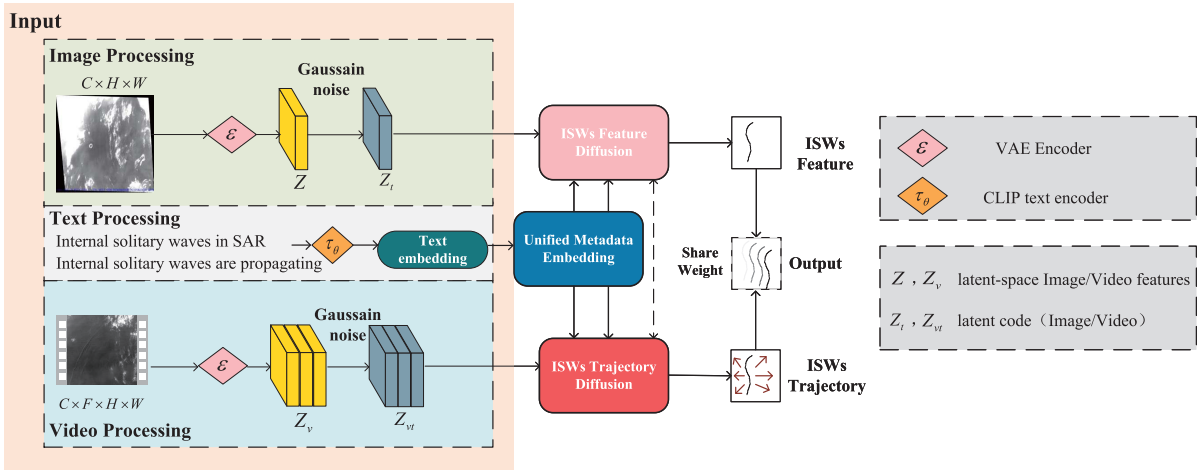


Fig. 3. Overall architecture of the proposed collaborative diffusion model. The framework consists of two core components: the ISWs feature diffusion (pink module) and the ISWs trajectory diffusion (red module). The ISWs feature diffusion model learns to generate the morphological features of ISWs from a single static image, while the ISWs trajectory diffusion model learns their propagation kinematics to form a temporal trajectory. These models collaborate and are constrained by a shared unified metadata embedding module, which ensures physical consistency between the generated features and trajectory. The VAE and CLIP encoders are used to map the input images/video and text into the latent space, respectively.

and fused with time-step embeddings, jointly serving as conditional guidance for the ISWs feature and trajectory diffusion models.

During the training phase, the ISW feature diffusion and ISW trajectory diffusion models are trained jointly, sharing weights between them. The low-rank adaptation (LoRA) technique is employed for fine-tuning. Finally, during the inference stage, the model generates the final visual sequence output.

Section IV-B will provide a detailed elaboration on unified metadata embedding, ISW feature diffusion, and ISW trajectory diffusion.

B. Unified Metadata Embedding

Currently, large-scale image datasets such as LAION-5B are typically accompanied by rich textual prompts [55]. In contrast, ISW's satellite imagery often lacks such descriptive annotations and is instead paired only with object detection boxes, segmentation masks, or categorical labels. This limitation poses substantial challenges for generating ISWs images using textual prompts, as models trained solely on segmentation masks or bounding boxes become difficult to optimize effectively. To address this issue, we observe that ISW images frequently come with metadata, such as timestamps, longitude, latitude, and other quantitative ISW-related variables. Based on this observation, we propose constructing a dataset in which each image x is associated with a textual description T and a set of metadata variables $m_j, j \in 1, \ldots, M$, where M denotes the number of metadata variables. In this way, the paper's aim is for the model to learn the data distribution $p(x|T, m)$. Thus, even when textual descriptions T are incomplete or missing, high-quality images can still be generated given accurate metadata m.

As shown in Fig. 4, each individual metadata value m_j is first projected into a high-dimensional feature space using

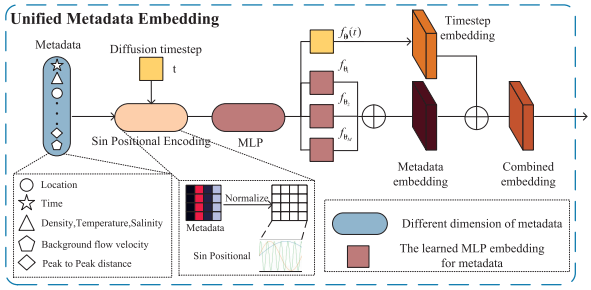


Fig. 4. Processing and embedding pipeline for metadata. Various types of raw metadata are first standardized through sinusoidal positional encoding and an MLP. Subsequently, these standardized metadata representations are aggregated into a single metadata embedding vector. This vector is finally combined with the timestep embedding to form the combined conditional embedding used to guide the diffusion model.

a sinusoidal positional encoding scheme, a technique proven effective in diffusion models. This projection is defined as

$$g(m, 2i) = \sin\left(m\Omega^{\frac{-2i}{d}}\right), g(m, 2i+1) = \cos\left(m\Omega^{\frac{-2i}{d}}\right). \quad (1)$$

Here, m represents the metadata, i denotes the index of the feature dimension, d is the dimension, and Ω is 10 000, a large constant. Initially, the metadata values are normalized into a range from 0 to 1, and subsequently projected using sinusoidal embedding. For each metadata variable, we encode the sinusoidally projected value with a distinct multilayer perceptron (MLP) to achieve a dimensional representation consistent with diffusion timestep t, as described by the following equation:

$$f_{\theta_j}(m_j) = \mathrm{MLP}\left[g(m_j, 0), \ldots, g(m_j, d)\right]. \quad (2)$$

Here, $f_{\theta_j}(m_j)$ denotes the learned MLP embedding corresponding to the jth metadata item. The final metadata vector, which we denote as $\mathbf{m_{eb}}$, is formed by summing these individual embeddings: $\mathbf{m_{eb}} = f_{\theta_1}(m_1) + \cdots + f_{\theta_M}(m_M)$, where $\mathbf{m_{eb}} \in$

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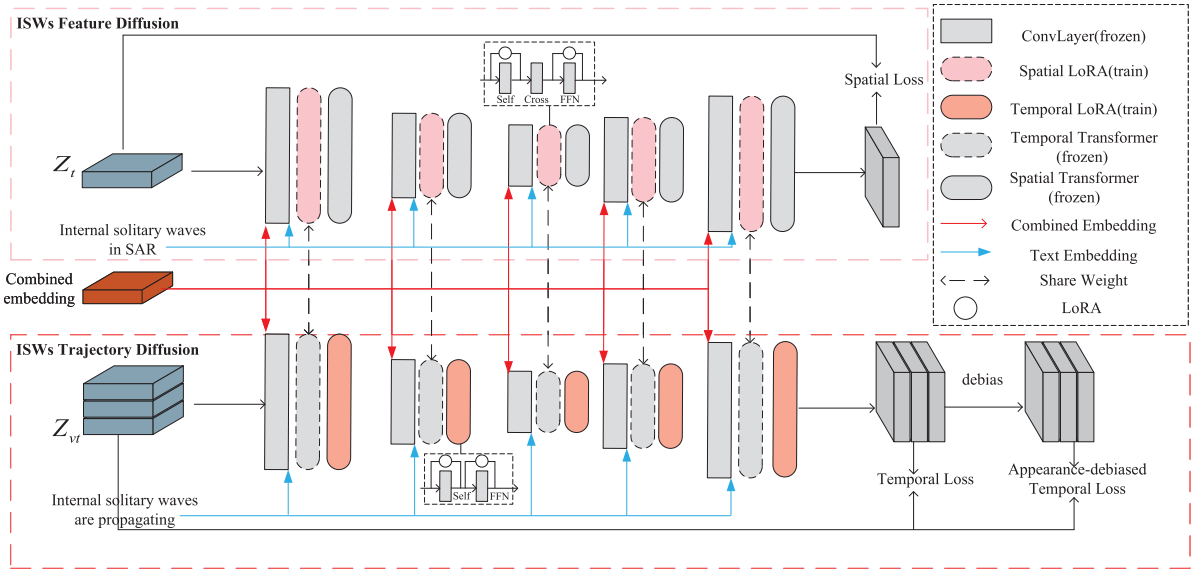


Fig. 5. Detailed structure of the dual-path diffusion model. The framework consists of two parallel U-Net models: the ISWs feature diffusion model (top) for learning appearance and the ISWs trajectory diffusion model (bottom) for learning motion. Both models adapt a pretrained, frozen backbone using LoRA. The feature model trains spatial LoRA modules (pink) to learn static appearance. The trajectory model shares and freezes these spatial LoRA weights while training temporal LoRA modules (orange) to learn motion. Finally, the loss function for the trajectory model undergoes a debiasing step to remove interference from appearance features.

$\mathbb{R}^D$, with D denoting the embedding dimension. Similarly, we use this same strategy to encode the diffusion timestep t. The t is first projected using $g(t)$ and then passed through its own dedicated MLP: $\mathbf{t_{eb}} = \text{MLP}_t(g(t)), \quad$ where $\mathbf{t_{eb}} \in \mathbb{R}^D$.

Subsequently, we process the primary visual and textual inputs. Stable diffusion-style variational autoencoder (VAE) encodes the input image $\mathbf{x} \in \mathbb{R}^{C\times H\times W}$ and the input video $\mathbf{y} \in \mathbb{R}^{F\times C\times H\times W}$ into latent representations $\mathbf{z} = \epsilon(\mathbf{x})$ and $\mathbf{z_v} = \epsilon(\mathbf{y})$. Gaussian noise is then added to these latent representations, resulting in the noised latent input $\mathbf{z_t}$ and $\mathbf{z_{vt}}$ for the two diffusion models. Simultaneously, the textual description T is encoded into a semantic embedding T' using a pretrained CLIP model.

Finally, the conditioning signals for the proposed framework are fully assembled. The diffusion timestep embedding, $\mathbf{t_{eb}}$, is added to the metadata vector to form the final numerical conditioning vector, $\mathbf{c} = \mathbf{m_{eb}} + \mathbf{t_{eb}}$. The framework is then trained by taking four inputs: the noised latent image $\mathbf{z_t}$, the noised latent video $\mathbf{z_{vt}}$, the textual embedding T', and the unified conditioning vector $\mathbf{c}$.

C. ISWs Feature Diffusion Model

To teach a massive, pretrained generative model the subtle visual fingerprint of ISWs without corrupting its powerful, preexisting knowledge, we employ LoRA [56]. LoRA allows us to specialize the model on ISW features in a parameter-efficient manner by freezing the original weights and training only small, low-rank matrices [57], formulated as

$$W' = W + \Delta W = W + AB \tag{3}$$

where $A \in \mathbb{R}^{d\times r}$ and $B \in \mathbb{R}^{r\times k}$ are learnable low-rank matrices, and $r \ll \min(d, k)$. Instead of updating the full weight matrix W, only the matrices A and B are trained, while W remains frozen. This significantly reduces the number of trainable parameters and memory requirements, making the adaptation process efficient and scalable.

The model's adaptation strategy involves the injection of LoRA modules within the model's spatial transformers, as shown in Fig. 5. We inject LoRA into the spatial self-attention and feed-forward layers to surgically modify the model's understanding of the characteristic shapes and surface textures of ISWs. Conversely, we do not inject LoRA into the cross-attention layers, thereby preserving the model's robust ability to interpret the conditioning vectors ($\mathbf{c}$,T'). This targeted adaptation allows the model to master the appearance of ISWs while retaining its general-purpose generative abilities.

Formally, the ISWs feature diffusion model is a U-Net denoted as ϵ_θ. During training, it takes four inputs: the noised latent $\mathbf{z_t}$, the integer timestep t, the numerical conditioning vector $\mathbf{c}$, and the text embedding T'.

The training objective for this path is a spatial reconstruction loss, focusing exclusively on perfecting the static visual appearance

$$\mathcal{L}_{\text{spatial}} = \mathbb{E}_{\mathbf{z}_0,t,\mathbf{c},T',\epsilon\sim\mathcal{N}(0,1)}\left[\|\epsilon - \epsilon_\theta\left(\mathbf{z}_t, t, \mathbf{c}, T'\right)\|_2^2\right] \tag{4}$$

where ϵ denotes the ground truth noise, and ϵ_θ is the predicted noise by the model. This loss forces the LoRA-enhanced layers to master the generation of high-fidelity ISW features from a noised latent $\mathbf{z}_t$ given the full set of conditions.

D. ISWs Trajectory Diffusion Model

To learn the physics-based motion of ISWs, we must prevent the model from "cheating" by relying on visual cues. The trajectory should be generated from the physical context ($\mathbf{c}, T'$),

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not just the appearance. We design a dedicated Trajectory Diffusion Model with two key innovations to enforce this.

First, the spatial LoRA modules are shared between the feature and trajectory models. A critical constraint is applied during backpropagation: the spatial LoRA modules only receive gradients from the spatial loss $\mathcal{L}_{\text{spatial}}$. They are effectively "frozen" against the temporal loss, receiving zero gradient from $\mathcal{L}_{\text{temporal-debiased}}$. This constraint forces the new, trainable temporal LoRA modules—which only receive gradients from the temporal loss—to focus exclusively on learning interframe relationships (the trajectory).

The model could be optimized with a standard temporal loss

$$\mathcal{L}_{\text{temporal}} = \mathbb{E}_{\mathbf{z_v},t,\mathbf{c},T',\epsilon\sim\mathcal{N}(0,1)}\left[\left\|\epsilon - \epsilon_\theta\left(\mathbf{z_{vt}}, t, \mathbf{c}, T'\right)\right\|_2^2\right]. \tag{5}$$

However, this is insufficient as the model may still use appearance to generate motion. To further disentangle trajectory from appearance, we adopt a decorrelation strategy.

The article defines the appearance feature $\mathbf{f}$ simply as the clean latent video tensor itself, $\mathbf{f} = \mathbf{z_v}$. Both the noise ϵ and the appearance latent $\mathbf{z_v}$ are 4-D tensors in the latent space, sharing the same shape ($\mathbb{R}^{F\times C'\times H'\times W'}$), where C', H', and W' are the latent dimensions. The debiasing is then formulated as an orthogonal projection, where $(\cdot)$ denotes the tensor inner product

$$\tilde{\epsilon} = \epsilon - \left(\frac{\epsilon\cdot\mathbf{z_v}}{\|\mathbf{z_v}\|^2}\right)\mathbf{z_v}. \tag{6}$$

The resulting $\tilde{\epsilon}$ is the component of the noise orthogonal to the clean appearance. The appearance-debiased temporal loss is then defined as

$$\mathcal{L}_{\text{temporal-debiased}} = \mathbb{E}_{\substack{\mathbf{z_v},t,\mathbf{c},T'\\ \epsilon\sim\mathcal{N}(0,1)}}\left[\left\|\tilde{\epsilon} - \epsilon_\theta\left(\mathbf{z_{vt}},\ t,\ \mathbf{c},\ T'\right)\right\|_2^2\right]. \tag{7}$$

For the ISW's diffusion system, training is performed jointly. The overall loss function is defined as the combination of the spatial component and the temporal component

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{spatial}} + \mathcal{L}_{\text{temporal-debiased}}. \tag{8}$$

During inference, both the trained spatial and temporal LoRA modules are injected into a diffusion backbone. Because we have successfully decoupled appearance and motion, we gain granular control: by adjusting the scaling of the respective LoRA weights, we can modulate the visual style and the dynamics, enabling a wide range of applications.

V. Experimental Evaluation

To comprehensively assess the performance of the proposed model, we design a series of experiments ranging from idealized numerical simulations to real-world satellite observations. The evaluation focuses on two primary aspects: the model's ability to reconstruct accurate ISW morphologies and their continuous spatiotemporal evolution, and its ability to generate realistic background visual fields. For background visual fields, we employ four standard metrics: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM), learned perceptual image patch similarity (LPIPS), and CLIP Score. PSNR is a classic metric that measures pixel-wise reconstruction fidelity by calculating the ratio between the maximum possible pixel value and mse between the generated and ground truth images. SSIM evaluates the perceptual similarity by comparing luminance, contrast, and structural information between two images. LPIPS computes the distance between deep features extracted from the generated and ground truth images using a pretrained deep neural network. CLIP Score measures the semantic consistency between the output image and the conditional text. It calculates the cosine similarity between the image and text embeddings in a shared latent space.

Beyond general perceptual assessments, we also explicitly evaluate the geometric and kinematic fidelity of the reconstructed ISWs. First, we digitize the ISW crestlines in both the generated and reference images. We then convert the pixel coordinates to meters using the scene's georeferencing data. For ISW morphology, we evaluate two core metrics: 1) the length L of wave crestline; and 2) the crest position error between the generated and ground-truth crests, which is summarized by the mean absolute error (MAE) and root mean square error (RMSE) of the shortest distances between them. For ISW kinematics, the article estimates the phase speed and propagation direction using two consecutive frames separated by a time interval of Δt. Specifically, we resample the crestline at time t_0 to have a near-uniform arc length and compute its Frenet frame. Each resampled point x on the t_0 crestline is matched to the nearest point y along the arclength of the crestline at time t_i

$$c = \frac{\|y - x\|}{\Delta t} \tag{9}$$

$$c_n = \frac{(y - x)\,n}{\Delta t} \tag{10}$$

$$\beta = (90^\circ - \text{atan2}\,(\Delta y, \Delta x)) \bmod 360^\circ. \tag{11}$$

As shown in Table I, L represents the length of the crest line, c represents the full speed, c_n represents the phase velocity, and β represents the propagation direction of the wave train. Here, the superscript "t" represents the observed value, and "g" represents the generated value. To reduce sensitivity to local curvature anomalies or mismatches, we robustly summarize the kinematic quantities across all crestline samples by calculating their median and interquartile range.

Finally, we use MAE and RMSE to quantify the error between generated and observed kinematic values and use directional error to assess the reasonability of the propagation direction. These geometry and physics-inspired metrics not only evaluate the plausibility of the generated ISWs but also validate their geometric fidelity and adherence to observed propagation characteristics.

A. Dataset

Unlike traditional methods requiring large-scale train/validation/test splits, our framework employs an event-specific single-sample customization paradigm based on the pretrained zeroscope_v2_576w model. The network is fine-tuned individually for each ISW event using one static reference image paired with an eight-frame video sequence.

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TABLE I
EVALUATE THE GEOMETRIC AND KINEMATIC REASONABILITY OF THE RECONSTRUCTED ISWS

	L^t (km)	L^g (km)	c^t (m/s)	c^g (m/s)	c_n^t (m/s)	c_n^g (m/s)	β^t (°)	β^g (°)
Numerical Data (average)	179.8	132.5	2.4 ± 0.2	2.4 ± 0.3	-2.3 ± 0.3	-2.4 ± 0.3	302.1	313.2
GHI (average)	166.6	172.0	1.6 ± 0.8	2.3 ± 1.3	-1.6 ± 0.9	-2.3 ± 1.3	291.9	291.9

* L represents the length of the wave crest line. c represents the full speed, c_n represents the phase velocity.
** β represents the propagation direction of the wave train. "t" and "g" represent truth and generation

To comprehensively ensure the robustness of our framework, our dataset comprises three categories, with each instance trained and evaluated independently as follows.

GHI Data: One reference image paired with its corresponding eight-frame continuous video. We independently verified 25 distinct GHI instances.

Numerical Simulation Data: One simulated reference image paired with its corresponding eight-frame continuous video. We independently verified 50 distinct simulation instances.

SAR Cross-Domain Data: Due to the lack of continuous SAR observations, we pair one SAR image with an eight-frame numerical simulation video. We independently verified 50 separate SAR instances. During preprocessing, rather than relying on rigid geographic bounding boxes, we extract image patches that explicitly encompass the main propagation features of the ISWs. These multisource inputs are then cropped, resized to a uniform resolution of 512×512 pixels, and normalized to facilitate cross-domain feature learning. Furthermore, temporal and spatial metadata were derived directly from the source imagery, while auxiliary physical parameters—including temperature, salinity, density, background field velocity, and flow velocity were integrated from HYCOM reanalysis data. The peak-to-peak distance was then analytically calculated.

B. Experimental Configuration

The LoRAs are trained using the AdamW optimizer with betas of 0.9 and 0.999, and an epsilon of $1e^{-8}$. We establish a learning rate of $5e^{-4}$, a weight decay of $1e^{-2}$, and apply a dropout rate of 0.1. The LoRA rank is configured to 32. For the ISW's trajectory diffusion training objective, the loss coefficients for both the original temporal loss and the appearance-debiased loss are equally weighted at 1. During data processing, we sample eight frames from each training video at a spatial resolution of 512×512 and a frame rate of 4 fps. The maximum number of training steps is specified as 300 for ISWs trajectory diffusion and 100 for ISWs feature diffusion. To explicitly define the convergence criteria and prevent overfitting, we employ an early stopping mechanism that halts the training process once the loss stabilizes and ceases to decrease. All experiments are implemented in PyTorch and conducted on a workstation equipped with an NVIDIA RTX 4090 GPU running Windows 10.

C. Evaluation With Numerical Simulation Data

We first evaluate the model in a highly controlled environment using numerically simulated ISW sequences. A mature numerical ocean model is employed to simulate the propagation of ISWs, from which image sequences are extracted. In addition, a single frame is randomly sampled from a total of 1009 timesteps to provide key physical features and associated ocean metadata. These are combined as input to the model.

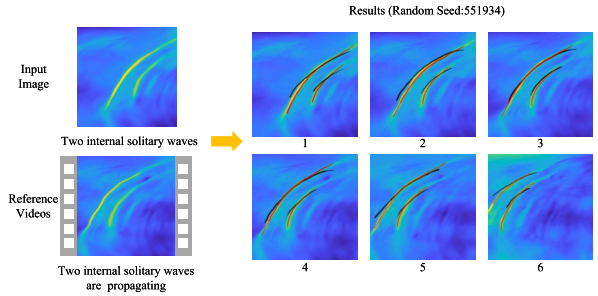


Fig. 6. Generated results of ISWs propagation. The model generates the subsequent six frames (1–6) based on an input image and a reference video. The random seed for this experiment is 551934.

As shown in Fig. 6, the model successfully captures the fundamental morphology of ISWs, and the reconstructed trajectories align well with the ground-truth numerical paths. Quantitative evaluation, summarized in Table II, further demonstrates the model's effectiveness. The model maintains strong reconstruction accuracy, with RMSE values ranging from 3.6 to 6.1 km under these idealized conditions.

D. Validation With GHI From FY-4B

To further evaluate the model's robustness and generalizability in real-world ocean environments, we conduct experiments using observational data from the FY-4B. Eight frames of real GHI imagery and one single-frame feature image are used as model input.

Fig. 7 presents the qualitative generation results. The top row shows the ground truth GHI sequence, while the middle row displays the proposed model's generated frames. The high visual fidelity is evident. The bottom row further visualizes the accuracy of the reconstructed trajectory (blue line) against the ground truth (green line). Moreover, the model successfully reconstructs the background scene with fine-grained consistency in texture and structure.

As shown in Table III, the model achieves high spatial accuracy, with the positional RMSE reaching as low as 1.92 km at the third reconstruction step. These results strongly support the model's high precision and reliability in challenging observational conditions, reinforcing its potential for continuous ISW observation and analysis.

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TABLE II

QUANTITATIVE EVALUATION ON THE ISW FEATURE GENERATION TASK. THE TABLE PRESENTS THE MODEL'S PERFORMANCE ON NUMERICAL SIMULATION DATA

STEP	LPIPS	SSIM	CLIP Score	PSNR (dB)	RMSE (km)	MAE (km)
1	0.568	0.680	25.580	11.400	6.044	3.279
2	0.554	0.673	25.600	11.440	5.096	3.988
3	0.555	0.665	25.580	11.370	5.053	3.206
4	0.585	0.659	25.570	11.350	4.428	2.758
5	0.593	0.662	25.130	11.360	3.640	1.863
6	0.627	0.674	25.200	11.240	5.729	4.497

* On the numerical simulation dataset, each step length is set to 10 minutes.

TABLE III

QUANTITATIVE EVALUATION ON THE ISW FEATURE GENERATION TASK. THE TABLE PRESENTS THE MODEL'S PERFORMANCE ON GHI DATA

STEP	LPIPS	SSIM	CLIP Score	PSNR (dB)	RMSE (km)	MAE (km)
1	0.481	0.611	21.130	14.610	3.018	2.522
2	0.461	0.645	20.320	16.350	2.513	1.908
3	0.420	0.689	20.820	17.810	1.916	1.558
4	0.453	0.662	20.380	17.420	2.539	1.749
5	0.418	0.702	20.290	18.160	2.310	1.831
6	0.396	0.710	20.720	18.570	5.077	3.008

* On the GHI dataset, each step length is set to 5 minutes.

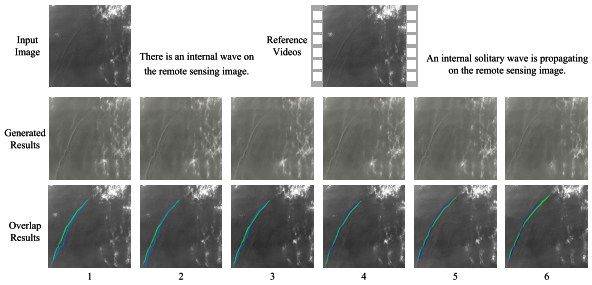


Fig. 7. Generated results of ISW propagation using GHI from the FY-4B. The model takes a single input image and a reference video to generate a six-frame visual sequence. The middle row ("generated results") shows the raw generated imagery, while the bottom row ("overlap results") visualizes the identified ISWs overlaid on the generation for enhanced clarity. The random seed for this experiment is 5155359.

TABLE IV

QUANTITATIVE EVALUATION ON THE ISW FEATURE GENERATION TASK. THE TABLE PRESENTS THE MODEL'S PERFORMANCE ON SAR

Number	LPIPS	SSIM	CLIP Score	PSNR (dB)
a	0.368	0.627	21.927	24.236
b	0.652	0.431	22.040	17.610
c	0.578	0.484	22.255	14.529
d	0.660	0.577	23.397	16.231

E. Application to Discontinuous SAR Observations

Building upon the previous experiments, we further explore the model's applicability to discontinuous real-world observations. Table IV shows quantitative evaluation on the ISW feature generation task. This experiment further highlights the model's applicability to discontinuous real-world observations. Specifically, we use a real SAR image as the source for feature extraction, while using numerical simulation sequences to drive the generation of ISWs trajectories.

Fig. 8 presents a comprehensive case study designed to validate the physical realism of the proposed model's generated results against real-world observations. The process begins with the MODIS image from the Terra satellite [Fig. 8(a)], which serves as the ground truth, clearly showing the presence of ISWs in the SCS [58]. The annotated lines track the propagation of the ISWs event over two consecutive days: the red line represents its position on April 21, 2007, and the yellow line its position on April 22. The corresponding raw SAR data from ENVISAT, capturing the same event, is shown in Fig. 8(b). Through the preprocessing pipeline, this data is refined into the clean, normalized input image seen in Fig. 8(c). Taking this single static image and dynamic visual sequence as input, the proposed model generates the multiframe temporal sequence depicted in Fig. 8(d). A qualitative analysis confirms a strong correspondence between the morphology and propagation patterns of the generated waves [Fig. 8(d)] and the actual ISWs observed in the MODIS imagery [Fig. 8(a)]. This result demonstrates that the proposed model accurately reflects the complex dynamics of ISW generation and evolution.

However, while the generated motion is coherent and the background textures align with the SAR input, this cross-domain experiment also reveals a notable limitation: the overall color tone is significantly distorted—dominated by an unnatural blue hue. To provide an in-depth analysis of the underlying causes, we attribute this color bias to two intertwined factors of semantic leakage [59]. First, it stems from the semantic prior embedded in the pretrained CLIP model, which strongly associates prompts containing the word "wave" with blue-ocean imagery derived from its large-scale training datasets. Second, there is an inherent feature entanglement within the cross-attention mechanisms of the latent diffusion model. During the cross-domain fusion, the network attempts to strictly disentangle the static speckle texture of the SAR reference from the smooth kinematic dynamics of the numerical simulation. Because these modalities possess vastly

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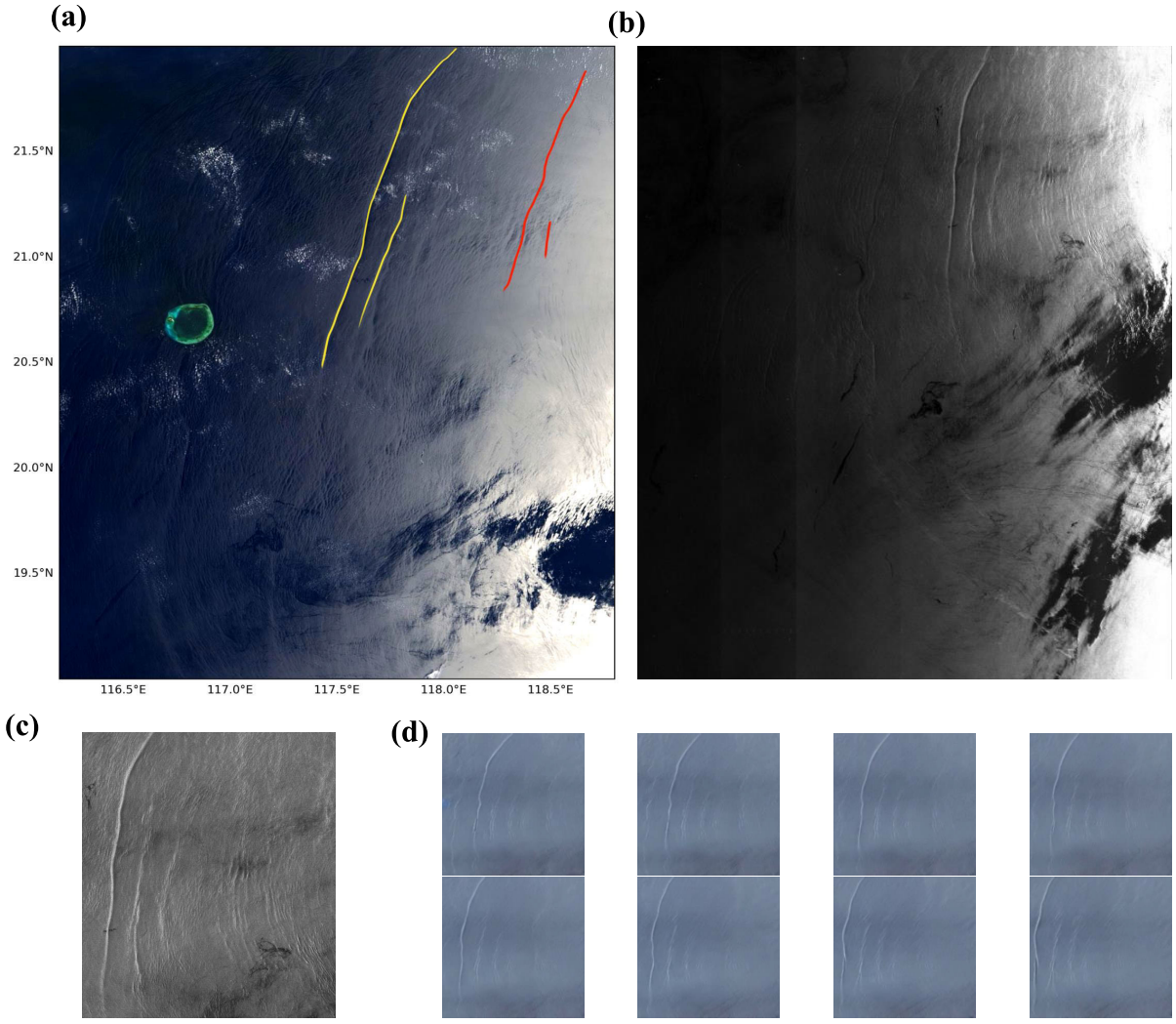


Fig. 8. (a) MODIS image from Terra. (b) Initial product from ENVISAT. (c) Processed image. (d) Model result. From the first row to the right, it, respectively, represents one frame. The random seed in this experiment is 3197403.

different pixel distributions, the attention maps designated to extract motion inadvertently capture the photometric style of the simulation data and the CLIP prior, transferring them onto the generated SAR wave crests.

To quantitatively assess this color distortion across different generation scenarios, we computed the color histogram correlation (CHC) and evaluated the mean pixel intensity shift. Our quantitative analysis reveals a stark contrast depending on the severity of the semantic leakage. In well-preserved instances, the generated sequences maintain a high structural and photometric fidelity, achieving an average CHC score of approximately 0.88 with a minimal intensity shift of 7%. However, in cases heavily affected by semantic leakage (where the unnatural blue hue dominates), the CHC score drastically degrades, approaching zero or even dropping to negative values. This near-zero or negative correlation mathematically confirms that the color distribution of the generated image has been completely decoupled from the original SAR domain. Concurrently, the mean pixel intensity in these distorted regions exhibits extreme shifts, ranging from 13% to over 50%.

This phenomenon, a severe misalignment between the learned visual prior and the actual input signal, is further illustrated across multiple SAR generation examples in Fig. 9. Despite the color distortion, these results consistently validate the model's core trajectory generation functionality in challenging cross-domain scenarios. Therefore, this series of SAR-based experiments not only highlights the model's robust capabilities but also clearly identifies its sensitivity to upstream semantic priors and cross-modal feature entanglement, offering a clear direction for future optimization.

F. Comparison With the KdV Equation

The behavior of ISWs can be described by various theoretical frameworks depending on the water depth conditions, including shallow water, deep water, and finite-depth theories. For shallow-water theory

$$\frac{L}{H} \gg 1, \quad \frac{h}{H} \sim O(1), \quad \frac{\eta L^2}{H^3} \sim O(1). \tag{12}$$

In this shallow water context, ISW propagation is often approximated by the Korteweg-de Vries (KdV) equation. In

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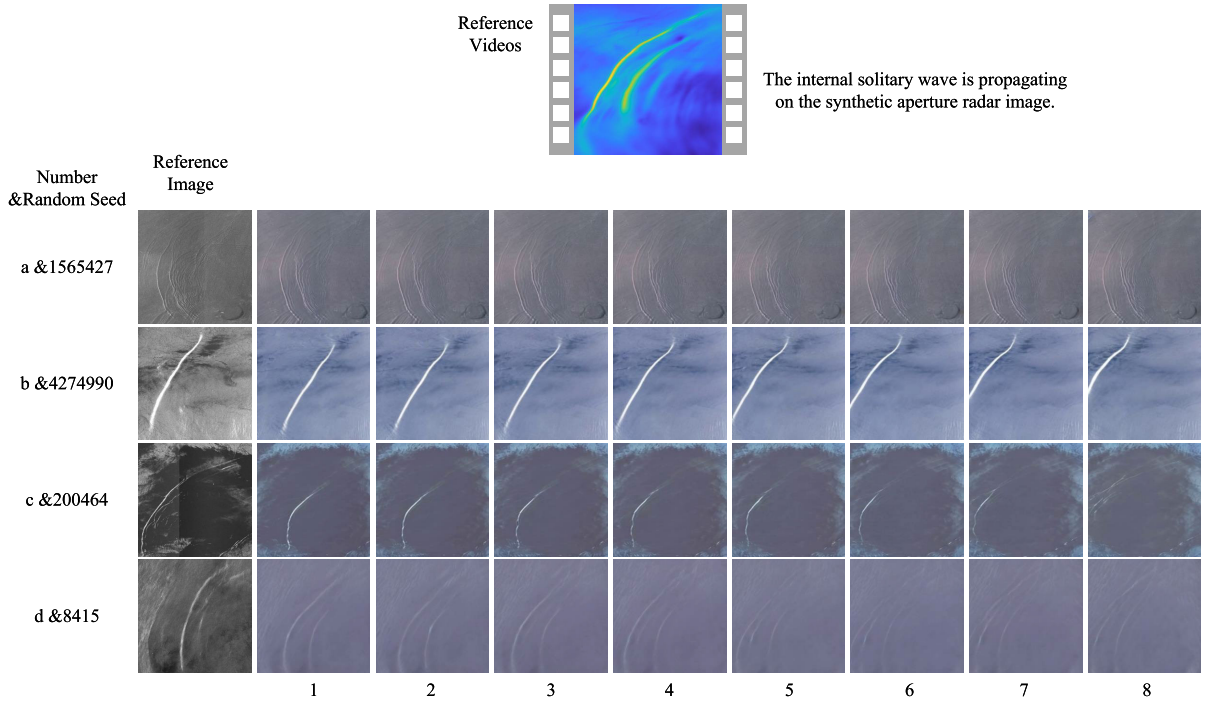


Fig. 9. Qualitative results of the model's generation on various SAR images. The figure displays four distinct examples (a–d), each initiated with a unique random seed. For each case, the model takes the first frame as input and, guided by the reference videos, generates the subsequent seven frames of ISW propagation.

this section, we compare the propagation results predicted by the theoretical KdV model against those generated by the proposed model.

According to KdV theory, the propagation velocity of an ISW is determined by its amplitude. The relationship is given by the formula

$$v = c_0 + \frac{\alpha}{3}\eta_0. \tag{13}$$

Here, c_0 represents the linear phase speed, α is the coefficient of nonlinearity, and η_0 is the initial amplitude of the ISW. A key advantage of this approach is that these parameters can be directly measured or estimated from satellite imagery, eliminating the need for unknown variables. For this experiment, we derived the following key parameters from satellite data: c_0 is 2.6082 m/s and α is 0.1932. η_0 is empirically set to 70 m. However, for large-amplitude waves, higher order nonlinearities become significant. Therefore, higher order models such as the extended KdV (eKdV) and modified KdV (mKdV) equations are also evaluated.

To evaluate performance, the RMSE was calculated at each time step to compare the actual satellite observations against the positions predicted by the KdV, eKdV, and mKdV models. All three theoretical models exhibited a clear, progressive increase in error over time. Specifically, across the six steps, the RMSE values grew from 1.34 to 7.82 for the KdV model, from 1.23 to 7.37 for the mKdV model, and from 1.12 to 6.75 for the eKdV model. In stark contrast, the proposed model was significantly less sensitive to the time series progression, maintaining a maximum RMSE of only 5.07 throughout the entire period, and its performance was significantly less sensitive to the time series progression.

The results conclusively show that, when compared against satellite observations, the ISW positions generated by the classic KdV equation exhibit progressively larger discrepancies over the 30-min period. Conversely, the trajectories generated by the proposed model remain robust, with the errors fluctuating within a much smaller and more stable range. This indicates that the model provides a more reliable and consistent generation of ISW propagation over time.

G. Visual Sensitivity Analysis of Metadata Constraints

To explicitly evaluate the impact of different physical metadata on the model's generation process and to validate our selection basis, we conducted a visual sensitivity analysis. Rather than relying solely on scalar error metrics, we visually compared the spatiotemporal propagation trajectories generated by the full collaborative diffusion framework against variants where specific metadata inputs were masked.

As illustrated in Fig. 10, the omission of specific metadata categories results in distinct and physically interpretable kinematic distortions. When the peak-to-peak distance is removed (Fig. 10 A), the model suffers from structural ambiguity in resolving the crest line, leading to the unphysical appearance of two simultaneously propagating ISWs. The removal of spatiotemporal metadata (Fig. 10 B) causes the model to lose its "spatial anchor." Given that geographic location serves as a critical proxy for regional bathymetry, its absence forces the generated ISWs into a generic, linear propagation pattern

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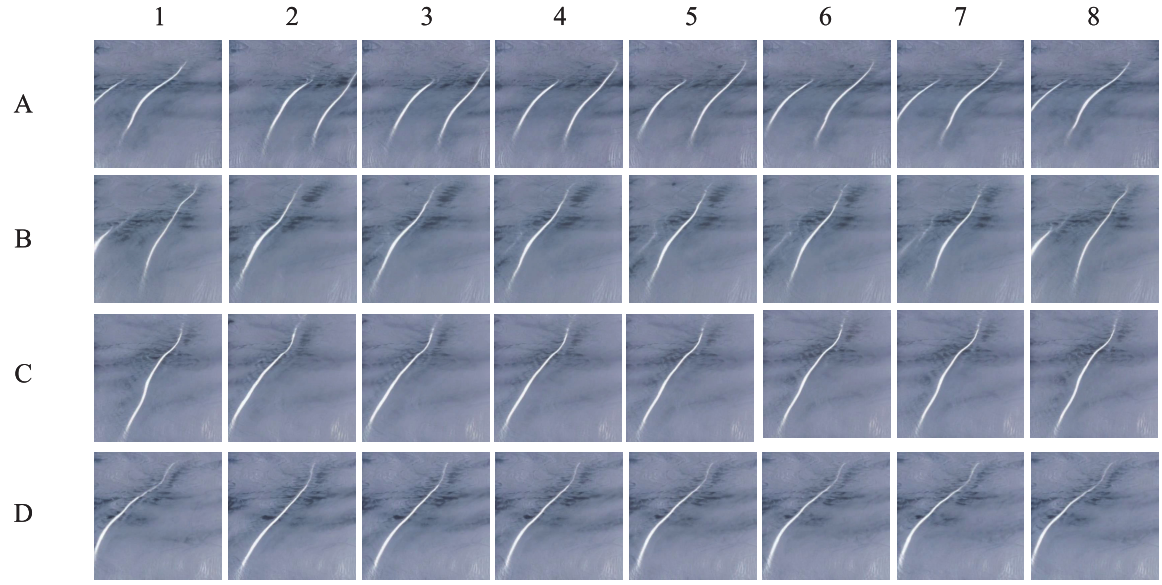


Fig. 10. Ablation of different metadata. Row A removes the peak-to-peak distance, Row B removes time and location, Row C removes the density, temperature, and salinity, and Row D removes background field velocity.

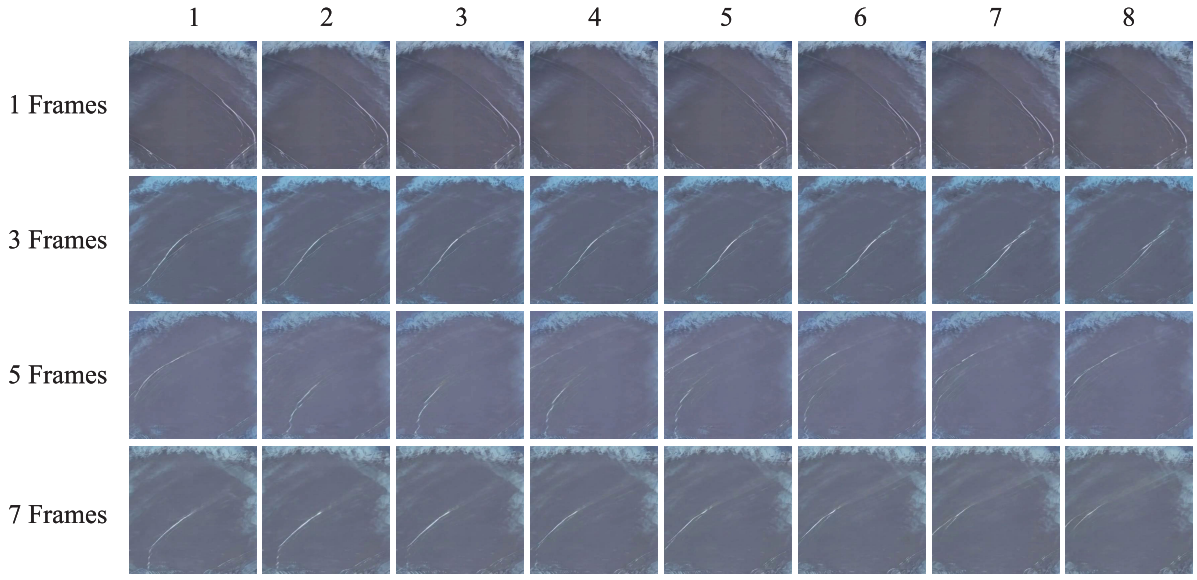


Fig. 11. Influence of different input video frame rates on the generation results.

that fails to capture the terrain-induced refraction and complex morphological bending characteristic of the study area.

Furthermore, omitting density, temperature, and salinity (Fig. 10 C) induces significant deviations in both propagation direction and phase speed, as the model can no longer account for density stratification effects. Finally, without the background flow velocity (Fig. 10 D), the generated wave packets exhibit retarded propagation and temporal incoherence, making it difficult to resolve the incremental displacement between successive frames.

This visual sensitivity analysis intuitively and conclusively demonstrates that the Unified Metadata Embedding is not a redundant feature, but a rigorous constraint mechanism. Each selected parameter quantitatively restricts the latent diffusion process, ensuring the generated sequences strictly adhere to the localized fluid dynamics.

H. Experimental Results Under Different Frames

This article systematically investigates the impact of temporal input length on SAR image synthesis by evaluating four frame-count configurations: 1, 3, 5, and 7 frames. Except for the number of input frames, the experimental settings remain consistent with the setup illustrated in Fig. 9(c). As depicted in Fig. 11, a single-frame input fails to resolve the initial position of the ISWs, leading to motion trajectories that deviate from physical regularities. While the three-frame configuration begins to align the propagation direction with expected physical laws, significant fidelity gaps remain. In contrast, the five-frame input yields a substantially complete reconstruction, although minor positional inaccuracies persist during the initial diffusion steps. Performance converges at the seven-frame mark, where the generated imagery achieves high

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realism and shows only negligible differences relative to the eight-frame baseline. These findings suggest that at least five frames are needed for stable generation, while seven frames already approach the performance of the eight-frame baseline. Nevertheless, the full eight-frame input still provides the most robust and structurally consistent results.

VI. Conclusion

In this study, we proposed and validated a novel collaborative diffusion framework that addresses the challenge of sparse satellite observations by reframing the propagation of ISWs as an end-to-end visual generation task. The core innovation of this framework is the successful disentanglement of the static appearance from the dynamic motion of ISWs and their embedding with physical information constraints, achieved through a dual-path collaborative mechanism.

Experimental results strongly demonstrate the effectiveness and robustness of the proposed framework. Across a series of tests ranging from idealized numerical simulations to real-world satellite imagery (SAR and GHI), the model not only generates visually realistic and physically plausible ISW propagation sequences but, most importantly, its generation error remains stable and does not accumulate over a 60-min time window. This characteristic represents a clear improvement over traditional KdV-based models. In addition, the proposed framework achieves an RMSE as low as 1.92 km on real GHI data.

This work has several important implications. First, it presents a new paradigm for data-driven ocean dynamic studies, proving the immense potential of generative models in solving the problem of spatiotemporal data scarcity. Second, the framework's ability to generate high-fidelity synthetic remote sensing visual sequences can serve as a valuable proxy dataset for training and evaluating models for other downstream tasks, such as ISW detection and segmentation.

Finally, this study also reveals some limitations of the current model. For instance, in cross-domain generation tasks, the model's semantic priors can lead to distortions in background color. Furthermore, achieving the optimal balance between the model's generative flexibility and the strict constraints of physical laws remains an area for exploration. Therefore, future work will focus on several key directions. First, we will aim to generate more realistic and context-aware ISW videos; for example, by conditioning the model on bathymetric data to account for topographic effects. Second, we will explore more advanced methods for integrating physical constraints. A highly promising direction is to incorporate the results from equation mining techniques, such as approximate governing equations discovered from data, as a powerful data-driven physical prior within the framework. This will serve to further enhance the model's generative fidelity and physical consistency, with the ultimate goal of enabling the model to not only replicate known trajectories but also to learn and generalize the underlying dynamics of ISW phenomena.

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